SYSTEM AND METHOD FOR HIGH-PRECISION FRACTIONAL NUMBER REPRESENTATION IN A DIGITAL COMPUTING SYSTEM

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Title: SYSTEM AND METHOD FOR HIGH-PRECISION FRACTIONAL NUMBER REPRESENTATION IN A DIGITAL COMPUTING SYSTEM

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Cross-Reference to Related Applications

This application claims the benefit of and priority to the portfolio of inventions described in the publication entitled "Be Water, Be Light," the contents of which are hereby incorporated by reference in their entirety. The system described herein is an improvement upon the high-cardinality parallel data bus and the parametric series number representation described therein.

Abstract

A system and method for representing fractional numbers in a digital computing system with high mathematical accuracy, avoiding the inherent imprecision of standard floating-point representations. A fractional number is represented as a composite N-gram data structure.

The integer part of the number is represented using a parametric series representation, while the decimal part is represented as a sum of scaled members from a known, convergent mathematical series, such as the series from the Basel problem. The representation of the decimal part is made highly efficient by encoding each term as a pair of indices into two predefined matrices: a matrix of series members and a matrix of multipliers. This "geo-indexing" approach allows for the transmission of the "ingredients" for a high-precision decimal number in a highly compressed format.

Background of the Invention

Modern computers represent decimal numbers using floating-point formats (e.g., float64). These formats are essentially a hack, representing a number as a binary integer significand and an exponent, which moves the decimal point. This system has inherent limitations in precision and can lead to rounding errors that accumulate over complex calculations. For applications requiring high mathematical fidelity, such as scientific simulation or geospatial analysis, these limitations are a significant drawback. There is a need for a method to represent and compute with fractional numbers in a way that is more mathematically accurate.

Summary of the Invention

The present invention provides a system for representing fractional numbers by separating them into their integer and decimal components and encoding each with a specialized, high-precision method for transmission or storage in a high-cardinality system.
The integer part of the number is represented using the parametric series method, where a "recipe" of (series_id, position, offset) is used. The decimal part is represented as a sum of scaled members of a known, convergent mathematical series, such as the Basel problem series $(1/n\hat{2})$. Each member of this sum is not transmitted by its value, but by a pair of indices that point to entries in two pre-shared matrices: a matrix of Basel problem members and a matrix of multipliers (k). This two-dimensional indexing, or "geo-indexing," method allows a specific scaled member (d = k * Basel@pos) to be represented by transmitting only the two index positions.

The integer representation and the multiple index pairs for the decimal part are grouped together into a composite N-gram of pixels or states in the high-cardinality interface. This

method allows a computer to transmit the "recipe" for a high-precision decimal number with far greater accuracy and efficiency than standard floating-point representations.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government, its military, and its Federal law enforcement agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a block diagram illustrating the composite N-gram structure for representing a high-precision fractional number.

FIG. 2 is a conceptual diagram of the "Two-Indices" or "Geo-Indexing" method for efficiently encoding the decimal part of a number.

Detailed Description of the Invention

Referring to **FIG. 1**, the structure for representing a high-precision fractional number (100) is shown. The number is first separated into its integer part (101) and its decimal part (102). The integer part is encoded using a parametric series representation (103), as previously described. The decimal part is represented as a sum of multiple scaled series members (104). Each of these components is encoded into one or more pixels or states in a high-cardinality interface, and they are grouped together to form a composite N-gram (105) that represents the entire fractional number. A mechanism, such as reserving a specific range of wavelengths or using relative positioning, is used to signify that the members of the N-gram belong together.

Referring to **FIG. 2**, the "Two-Indices" or "Geo-Indexing" method for encoding the decimal part is illustrated. To represent a scaled member of the decimal sum, d = k * Basel@pos, the system does not transmit the values of k or pos directly. Instead, it uses two pre-shared

matrices. A 'Matrix of Multipliers' (201) stores a range of possible k values. A 'Matrix of Basel Members' (202) stores a range of pre-calculated members of the series. The scaled member is represented by a pair of indices: an index (203) into the multiplier matrix to select k, and an index (204) into the Basel member matrix to select the position. This pair of indices (205) is then encoded into a single state in the high-cardinality interface. This method is highly efficient, as it replaces the transmission of two potentially large numbers (k and pos) with two small index numbers.

Claims

What is claimed is:

1. A method for representing a fractional number in a digital system, comprising the steps of:
    - separating the fractional number into an integer part and a decimal part;
    - encoding the integer part using a parametric series representation;
    - representing the decimal part as a sum of one or more scaled members of a pre-determined convergent mathematical series; and
    - combining the encoded integer part and the representation of the decimal part into a composite data structure.
2. The method of claim 1, wherein the step of representing the decimal part further comprises:
    - for each scaled member of the sum, encoding it as a pair of indices into at least two pre-shared matrices, one containing multipliers and one containing members of the convergent series.
3. The method of claim 2, wherein the convergent mathematical series is the Basel problem series.
4. The method of claim 1, wherein the composite data structure is an N-gram of states in a high-cardinality, parallel data interface.
5. A system for representing a fractional number, comprising:

- a module for separating a fractional number into an integer part and a decimal part;
- a first encoding module for encoding the integer part as a parametric series representation;
- a second encoding module for representing the decimal part as a sum of scaled members of a convergent series, wherein each scaled member is encoded as a pair of indices into pre-shared matrices; and
- a module for combining the outputs of the first and second encoding modules into a composite N-gram data structure for transmission or storage.

6. The system of claim 5, wherein the pre-shared matrices comprise a matrix of multipliers and a matrix of series members.

Drawings

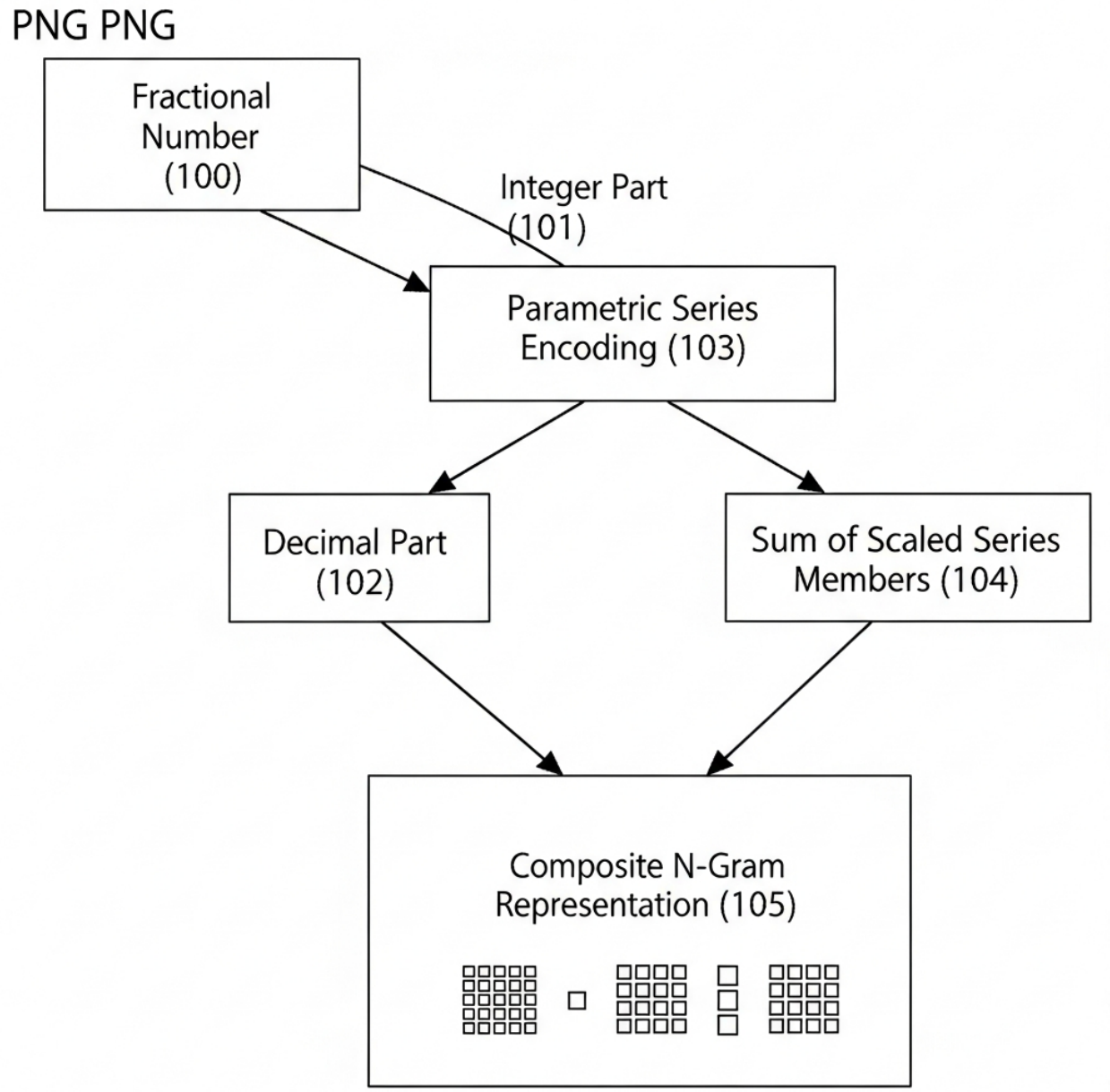


Figure 1: Block diagram of the composite N-gram structure.

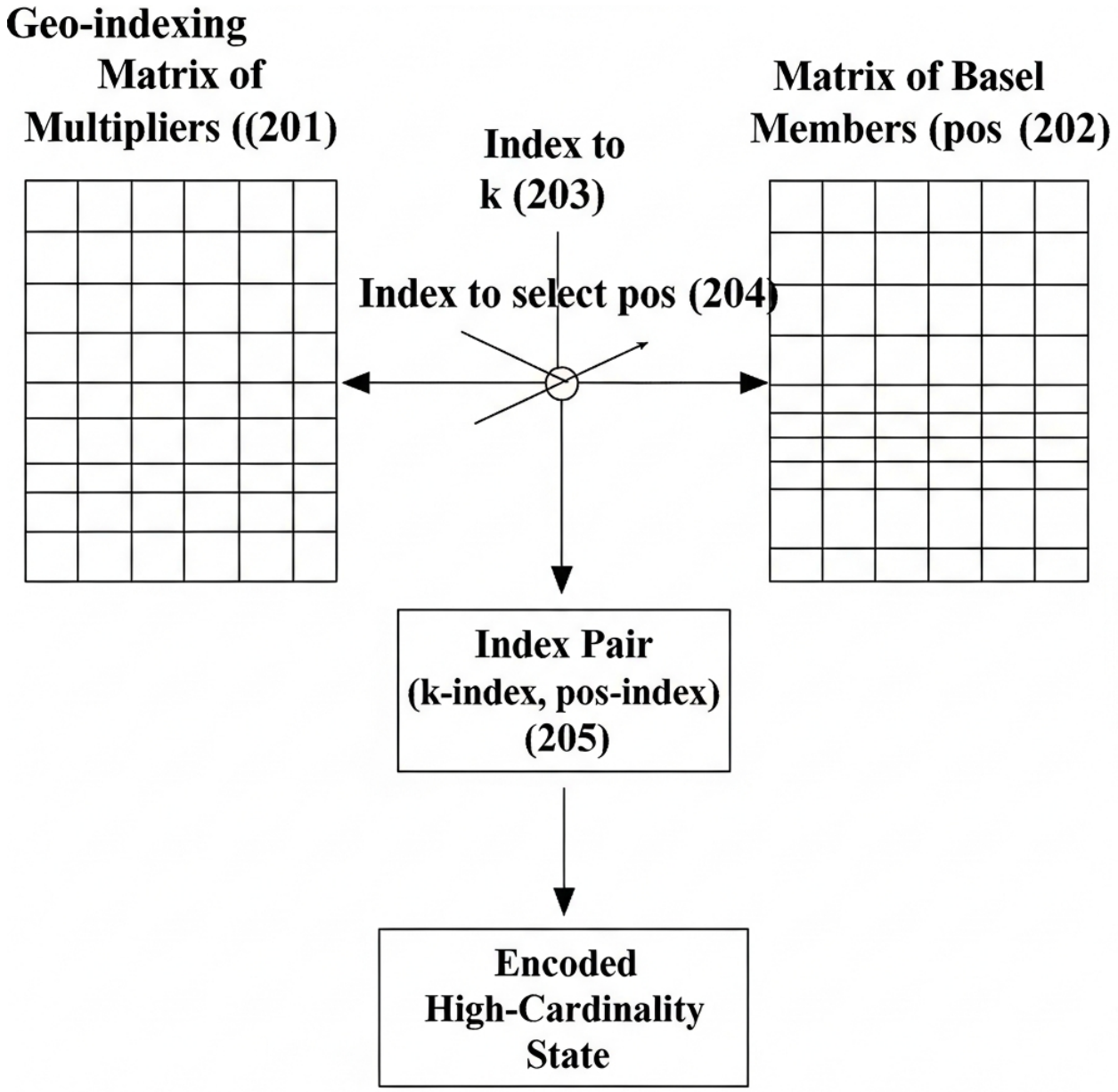


Figure 2: Conceptual diagram of the "Two-Indices" method.